

[5254]-533

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Neat diagrams must be drawn wherever necessary.
- 2) Figure to the right indicate full marks.
- 3) Use of electronic calculator is allowed.
- 4) Assume suitable data, if necessary.

**Q1) a)** A four cylinder vertical engine has cranks 150 mm long. The cylinders are spaced 200 mm apart. Mass of reciprocating parts of 1<sup>st</sup>, 2<sup>nd</sup> and 4<sup>th</sup> cylinders are 50 kg, 60 kg and 50 kg respectively. Find the reciprocating mass of the 3<sup>rd</sup> cylinder and relative angular positions of the cranks to achieve complete primary balance. [6]

**b)** Determine the expression for natural frequency of the system shown in Fig. 1 [4]

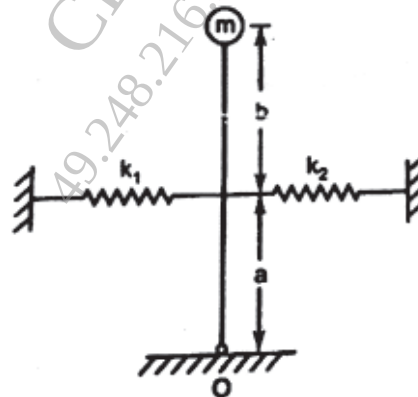


Fig. 1 (Q. 1b)

OR

**Q2) a)** A shock absorber is to be designed so that its overshoot is 10% of the initial displacement when released. Determine the damping factor. Also find the overshoot if the damping factor is reduced to 50%. [6]

**b)** Explain the terms Static Balancing and Dynamic Balancing. [4]

P.T.O.

**Q3) a)** A single cylinder vertical petrol engine of total mass 320 kg is mounted on a steel chassis and causes a vertical static deflection of 2 mm. The reciprocating parts of the engine have a mass of 24 kg and move through a vertical stroke of 150mm with SHM. A dashpot attached to the system offers a resistance of 490 N at a velocity of 0.3 m/s. Determine : [6]

- i) the speed of driving shaft at resonance
- ii) the amplitude of steady state vibrations when the driving shaft of the engine rotates at 480 rpm.

**b)** Define the following terms : [4]

- i) Damping coefficient
- ii) Critical damping coefficient
- iii) Damping factor
- iv) Logarithmic decrement

OR

**Q4) a)** A horizontal spring mass system with coulomb damping has a mass of 5 kg attached to a spring of stiffness 980 N/m. If the coefficient of friction is 0.25, calculate : [6]

- i) the frequency of free oscillations
- ii) the number of cycles corresponding to 50% reduction in amplitude if the initial amplitude is 5 cm
- iii) time taken to achieve this 50% reduction

**b)** Write a short note on Forced vibrations due to reciprocating unbalance.[4]

**Q5) a)** Find the natural frequencies of the system shown in Fig. 2. [12]

$$m_1 = 10 \text{ kg}, m_2 = 12 \text{ kg}$$

$$r_1 = 0.10 \text{ m}, r_2 = 0.11 \text{ m}$$

$$k_1 = 40 \times 10^3 \text{ N/m}$$

$$k_2 = 50 \times 10^3 \text{ N/m}$$

$$k_3 = 60 \times 10^3 \text{ N/m}.$$

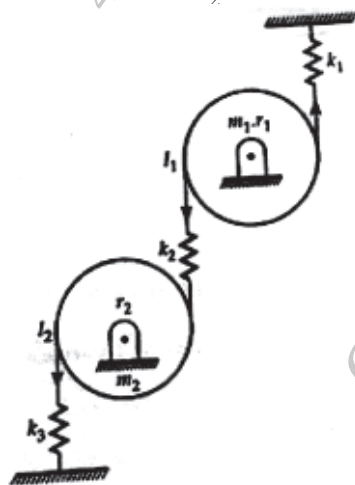


Fig. 2 (Q. 5 a)

b) Define the following terms : [4]

- i) Zero frequency                      ii) Node point

OR

**Q6) a)** Find the natural frequencies and mode shapes for the torsional system shown in Fig. 3. Assume  $J_1 = J_0$ ,  $J_2 = 2J_0$  and stiffness for each spring as  $k_t$ . [12]

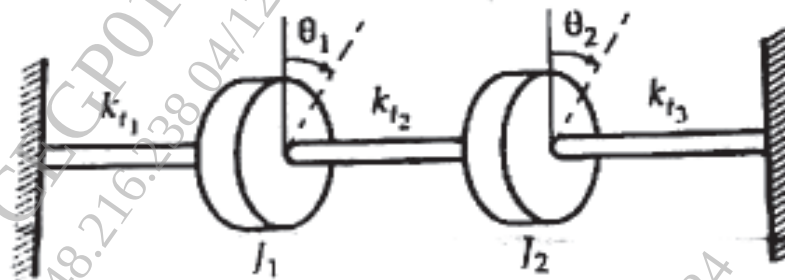


Fig. 3 (Q. 6a)

b) Explain the concept of torsionally equivalent shaft. [4]

**Q7) a)** An accelerometer has a suspended mass of 0.01 kg with a damped natural frequency of vibration of 150 Hz. It is mounted on an engine running at 6000 rpm and undergoes an acceleration of 1 g. The instrument records an acceleration of  $9.5 \text{ m/s}^2$ . Find the damping constant and the spring stiffness of the accelerometer. [8]

b) Write a short note on prediction of vibration failure using time and frequency domain analysis of vibration signals. [8]

OR

**Q8) a)** For finding vibration parameters of a machine running at 260 rpm, a seismic instrument is used. The natural frequency of the instrument is 7 Hz and the recorded displacement is 6 mm. Determine the displacement, velocity and acceleration of the vibrating machine assuming no damping. [8]

b) Write a short note on : [8]

- i) FFT analyzer  
ii) Condition monitoring of machines

**Q9) a)** Determine the sound power level of a source generating [8]

- i) 0.5W                      ii) 1.5W  
iii) 2.2 W                  iv) 3W of sound power

b) Explain the following terms: [10]

- i) Wavelength
- ii) Velocity of sound
- iii) Decibel scale
- iv) Sound power level
- v) Sound pressure level

OR

**Q10)a)** Define the following terms : **[6]**

- i) Reflection coefficient                      ii) Absorption coefficient
- iii) Transmission coefficient

b) Draw and explain the main components of human hearing mechanism.[6]

c) Show that if the sound pressure is doubled, the sound pressure level increases by six decibels. [6]

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